

MY CYBER ACADEMY

Unit U01

CREST Practitioner Intrusion Analyst (CPIA)

Course Introduction & Soft Skills / Incident Handling

The analyst mindset, the IR lifecycle, and the law and threat models that frame every engagement.

GROUP A | SOFT SKILLS | INCIDENT HANDLING

© Billy Forrest

Fourteen units, six knowledge groups, one certification

Knowledge Group	Focus	units
A · Soft Skills & Incident Handling	Engagement, chronology, law, threat	1
B · Core Technical Skills	Protocols, tools, fingerprinting, crypto, host analysis	2–4
C · Background Info Gathering & OSINT	WHOIS, DNS, open-source investigation	5
D · Network Intrusion Analysis	Capture, logs, C2, exfiltration, attacks	6–8
E · Analysing Host Intrusions	Acquisition, Windows FS, registry, memory, malware	9–11
F · Reverse Engineering Malware	Anti-RE, PE format, obfuscation, behaviour	12–14

© Billy Forrest

UNIT OBJECTIVES

By the end of this unit you will be able to

- **Describe the CPIA exam** its six knowledge groups and how this course maps to them
- **Explain the analyst role** the traits and experience that make a good analyst, and the impact of bias
- **State the IR lifecycle** suspect, acquire, analyse, report, deploy countermeasures
- **Explain the cyber kill chain** the seven phases of a targeted intrusion
- **Apply law and compliance** to incident handling and evidence
- **Use the TAME model** and risk treatment options in threat assessment

© Billy Forrest

What makes a good analyst?

- **Flexibility** adapt to changing circumstances
- **Logical and independent mind** question assumptions and reason clearly
- **Attention to detail** the small clues carry the investigation
- **Objectivity and sensitivity** handle confidential information with care
- **Concentration and patience** long analysis units, careful work
- **Confidence in your own judgement** defensible conclusions

© Billy Forrest

Where analyst skills come from

Background	Transferable skill
IT / troubleshooting	Diagnosing problems methodically
Networking	Setup, configuration and protocol awareness
IT hardware	Storage identification, component handling
Software engineering	Low-level computing, scripting
Law enforcement	Continuity, direction, human evidence sources

© Billy Forrest

Analyst bias

- **Conscious and unconscious bias** built through upbringing, culture, education and occupation
- **Cognitive bias** systematic errors in thinking that distort judgement
- **Forensic confirmation bias** pre-existing beliefs and motives shape how evidence is gathered
- **The risk** prejudicial information — ethnicity, criminal record — compromises impartiality
- **The guard** recognise and actively counter bias in your analysis

© Billy Forrest

DISCUSSION

Why must an analyst actively guard against confirmation bias?

Think of a case where you already believe you know who the intruder is. How does that belief change what you look for, and what you dismiss?

Confirmation bias can steer an entire investigation toward a premature conclusion.

© Billy Forrest

Benefits and utility of incident response

- **A framework for incidents** reduces errors and prevents loss of critical forensic data
- **A clear path to the cause** establishes how the incident happened
- **Legal and stakeholder notification** all parties informed when required
- **Reduced reputational damage** effective management of the incident
- **Rapid, systematic recovery** business impact kept to a minimum

© Billy Forrest

Preparation before an incident

- **Establish an IR team** roles and responsibilities agreed in advance
- **Conduct threat analysis** know the likely threat actors
- **Outline response guidelines** and procedures for external communication
- **Communicate with personnel** and understand the relevant policies
- **Train employees** and test the IR plan
- **Capture lessons learned** improve the plan after every exercise

© Billy Forrest

The IR lifecycle

1. **Suspect a compromise**
   human or machine detection of an anomaly
2. **Acquire data**
   memory dumps, packet captures, logs, disk images
3. **Analyse data**
   correlate and reconstruct the timeline
4. **Report**
   a forensic, structured summary of the compromise
5. **Deploy countermeasures**
   prevent recurrence and harden the posture

© Billy Forrest

The cyber kill chain

1. **Reconnaissance**
   research, identify and select targets
2. **Weaponisation**
   pair remote-access malware with an exploit into a deliverable
3. **Delivery**
   transmit the weapon to the target
4. **Exploitation**
   the weapon executes and runs code
5. **Installation**
   a backdoor is installed for persistence
6. **Command and control**
   the code connects back for instructions
7. **Actions on objective**
   the attacker pursues their aims

© Billy Forrest

QUIZ

Activity: map the kill chain

- A user opens a PDF that drops a backdoor — which kill chain phase?
- The malware phones home for instructions — which phase?
- The attacker pivots to steal data — which phase?
- A spear-phishing email is crafted for a named victim — which phase?
- Order the seven phases for a ransomware intrusion.

Recon → Weaponise → Deliver → Exploit → Install → C2 → Act on objective.

© Billy Forrest

Incident chronology

- **Reconstruct the timeline** order events from all evidence sources
- **Correlate sources** network logs, host logs, captures, memory
- **Establish a chain of events** what happened, when, and in what order
- **Feed the bigger picture** cross-reference with historical reports
- **Enable management metrics** and shared threat intelligence

© Billy Forrest

Law, compliance and evidence

- **Engagement lifecycle** scope, legality and record-keeping
- **Applicable legislation** Computer Misuse Act, Data Protection Act, GDPR
- **Evidence integrity** do not change data; preserve the audit trail
- **Independent review** a third party should reproduce your result
- **Confidentiality** sensitive data handled appropriately

© Billy Forrest

Risk ownership and the TAME model

- **Risk ownership** each risk is owned by a named stakeholder
- **Documentation** supports ISO 27001, ISO 9001 and other certifications
- **TAME model** Threat Assessment Model for Electronic payment systems
- **Four stages** assessment scope; threat agent & vulnerability analysis; scenario construction & system modelling; evaluation
- **Scalable** built for payment systems, extensible to enterprise networks

© Billy Forrest

STRIDE for the risk owner

STRIDE	Threat class	The owner asks
S	Spoofing	Can someone claim another identity in our logs?
T	Tampering	Can evidence be altered before the report?
R	Repudiation	Can a party deny the action, and do we hold proof?
I	Information disclosure	Who can read the data, at rest and in transit?
D	Denial of service	What stops the investigation at the worst moment?
E	Elevation of privilege	Can a low privilege account reach the evidence?

© Billy Forrest

Risk treatment options

Option	Meaning	Example
Acceptance	No action necessary	Low-risk, low-impact residual risk
Avoidance	Risk is no longer present	Stop the activity entirely
Reduction	Risk is less of a threat	Add controls to lower likelihood
Transference	Risk is someone else's problem	Insurance, outsourcing

© Billy Forrest

Attribution: state vs non-state attackers

Attacker	Characteristics
State	Government / military / intelligence; very sophisticated; large budget; long-running campaigns; target national infrastructure
Non-state	Individuals / organised groups / terrorists; lower sophistication; small or no budget; often profit or thrill driven

© Billy Forrest

Group A at a glance

Ref	Topic	Core idea
A1	Engagement lifecycle management	Prepare, detect, acquire, analyse, report, countermeasure
A2	Incident chronology	Reconstruct and correlate the timeline
A3	Law & compliance	Legal framework, evidence integrity, confidentiality
A5	Threat assessment	TAME model, risk ownership, risk treatment

© Billy Forrest

KEY TAKEAWAYS

- **The analyst role** demands objectivity, attention to detail and active bias management
- **IR is a lifecycle** suspect → acquire → analyse → report → countermeasure
- **The kill chain** seven phases — recon to actions on objective
- **Law frames the work** evidence integrity, confidentiality and legality
- **Threats are assessed** with the TAME model and explicit risk treatment

© Billy Forrest

QUIZ

confirmation quiz (Group A)

- Which IR phase collects memory dumps and packet captures?
- What is the key difference between a state and a non-state attacker?
- Risk transference means the risk is:
- Which kill chain phase follows 'installation'?
- Why does confirmation bias threaten an investigation?

Answers in the confirmation quiz — aim for 60 percent before leaving.

© Billy Forrest

NEXT SESSION

Next unit: Core Technical Skills I

- IPv4 and IPv6 addressing, TCP, UDP and ICMP
- Application-layer protocols: HTTP(S), SMTP, DNS, FTP, SMB, SNMP
- The OSI model and how the web and email work
- Network architectures and common tool classes

Reading: Knowledge Group B source material.

© Billy Forrest